

AMEY SANTOSH RANE

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SUMMARY

Software engineer with 3+ years across **Mobile, AI/LLM, full-stack, and test automation**. Grew from Android developer to **AI & Mobile Systems Lead** on an NIH-funded clinical research platform, architecting **React Native + Firebase + OpenAI** pipelines on **MVVM modular architecture**. Prior SDET work at **Abbott**.

PROFESSIONAL EXPERIENCE

Technical Lead | *University at Buffalo Foundation*

Buffalo, NY | DEC 2025 – Present

- Lead engineering direction of a cross-platform **React Native** healthcare app in active NIH-funded clinical research; coordinate delivery across mobile, backend, and research teams and mentor engineers on architectural standards.
- Architected end-to-end **AI pipeline** integrating **OpenAI GPT, Firebase Firestore, and Cloud Functions** for personalized conversational support, behavioral nudges, and real-time participant engagement.
- Established **MVVM-based modular architecture** with clear separation between **UI, view-models, and service layers**, standardized reusable hooks, navigation patterns, and Firebase data access for scalable delivery.
- Driving **ML initiatives** on wearable sensors: designing IMU data collection and labeling pipelines on a smart-ring device for hand-gesture classification using feature-engineered and sequence-model classifiers.
- Built **push notifications, GPS services, and offline-tolerant sync for field studies**, authored integration tests covering service-to-service communication across distributed components.

React Native Mobile Developer

AUG 2025 – DEC 2025

- Led migration from native Android to **cross-platform React Native**, re-architecting core modules into reusable, testable components backed by Firebase Auth and Firestore real-time sync; designed gamification (XP, streaks, rewards) and community chat features.

Android Feature Developer

MAY 2025 – AUG 2025

- Built native Android features (behavioral tracking, check-ins, activity modules) in **Java/Kotlin** using Activities, Fragments, Services, and lifecycle-aware components; debugged via ADB/logcat across emulators and physical devices.

Software Engineer in Test | *Abbott Diabetes Care*

Alameda, CA | MAY 2023 – AUG 2023

- Built automated UI and integration test frameworks for **Android and iOS** CGM medical-device apps using **Java, Appium, Selenium, Cucumber/BDD** under **IEC 62304 / ISO 13485** regulated standards.
- Authored unit, integration, and UI test suites validating mobile features against wearable telematics and RESTful backends; improved regression reliability through reusable CI test utilities.
- Partnered with cross-functional engineers on **V&V workflows, JIRA defect triage, and SOP documentation** supporting long-term framework adoption.

Full-Stack Software Engineer | *Techno Win IT Infra Pvt Ltd*

Mumbai, India | OCT 2021 – MAY 2022

- Built full-stack web applications with **ASP.NET (MVC), React.js, and MySQL** plus RESTful APIs for enterprise reporting and internal analytics tools.
- Developed **Python and SQL** data ingestion, validation, and transformation pipelines; applied relational schema design and query optimization for analytics accuracy.
- Configured **Azure CI/CD; supported production debugging, data-quality issues, and ETL** improvements in a fast-paced startup environment.

EDUCATION

M.S in Computer Science, Binghamton University (SUNY), GPA 3.5/4.0

New York, USA | AUG 2022 – MAY 2025

B.E in Electronics & Communication, University of Mumbai, GPA 3.2/4.0

Mumbai, MH, IN | AUG 2016–OCT 2020

TECHNICAL SKILLS

- **Languages:** JavaScript (ES6+)/TypeScript, Java, Kotlin, Swift, Python, C#, SQL, Gherkin, HTML/CSS
- **Mobile:** React Native, Android (Java/Kotlin), iOS (Swift), Firebase (Auth/Firestore/Cloud Functions)
- **AI / ML:** OpenAI & Claude APIs, LLM Integration, AI Agents, IMU Sensor Data, Data Labeling
- **Full-Stack & Data:** React.js, ASP.NET, Django, Node.js, MySQL/MS-SQL, pandas, NumPy
- **Test Automation:** Appium, Selenium, Cucumber/BDD, XCTest, UIAutomator2, TDD
- **Cloud / DevOps:** AWS (EC2, S3), Azure, Docker, GitHub Actions, Jenkins, Git, Bitbucket, JIRA, Confluence
- **Certifications:** AWS Certified Cloud Practitioner; Python-Django Full Web Development

PUBLICATIONS & PROJECTS

Health Monitoring & Safety System for Alzheimer's Patients IRJET, May 2020 [[Paper](#)].

- IoT + Raspberry Pi system with a React Native / Firebase companion app for real-time caregiver alerts and secure data sync.

LeetCode 2.0(Open-source contributor) [[GitHub](#)].

- Engineered automated metadata parsing and categorization for LeetCode submissions synced to GitHub, organized by problem domain and difficulty.

OmniSch: Benchmark for Schematic Understanding & Spatial Netlist Graph Construction (Co-author) [[Paper](#)].

- First comprehensive benchmark assessing **Large Multimodal Models (LMMs)** on electronic schematic interpretation and graph-structured spatial reasoning.